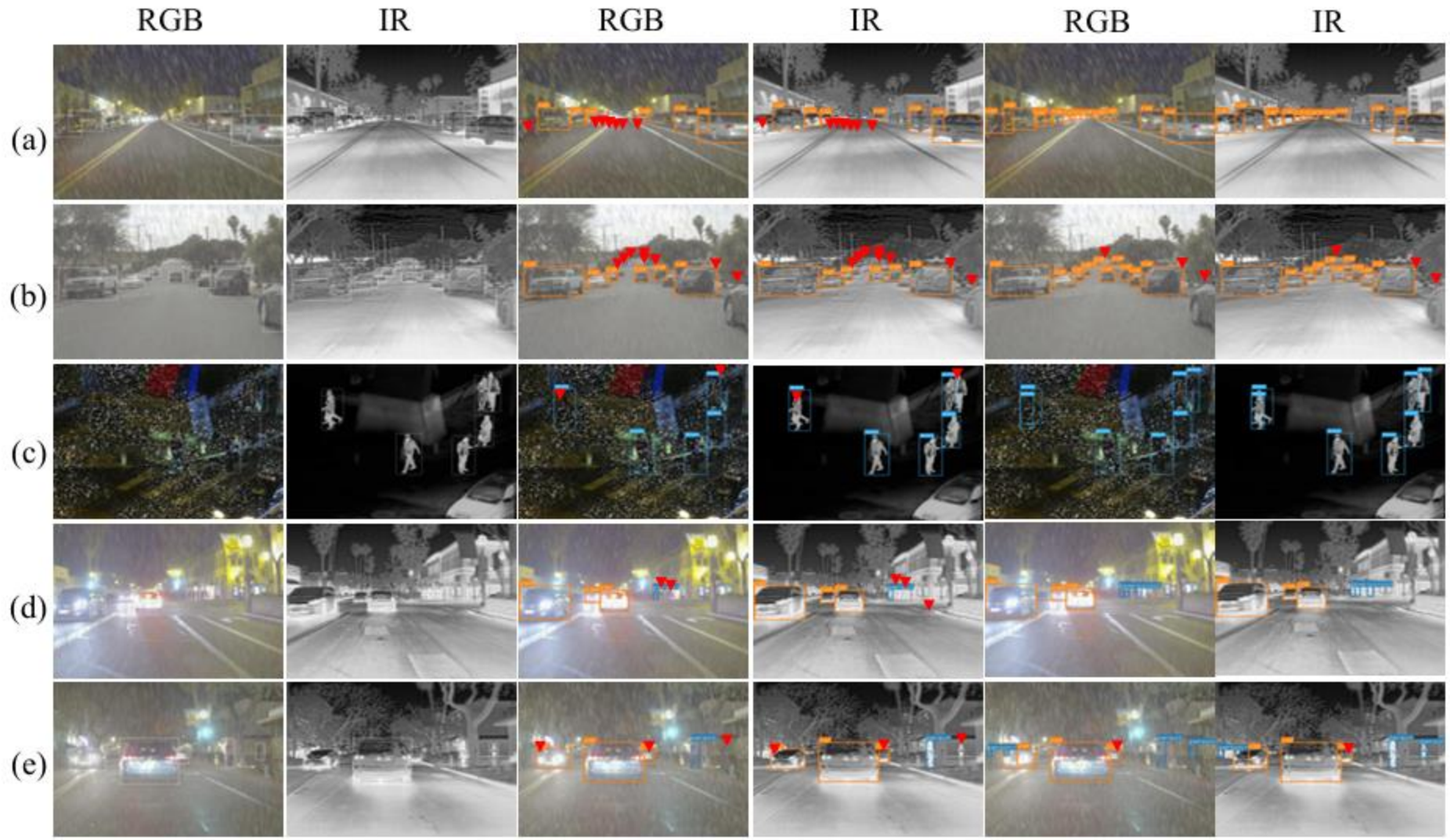
OBJECT DETECTION USING CYCLE-GAN GENERATED PSEUDO RGB THEMAL IMAGES





Source: CFMW: Cross-modality Fusion Mamba for Robust Object Detection under Adverse Weather. Li et al. 2025

- Deep object classification in low resolution LWIR imagery via transfer learning. Abbot et al. 2017
- Borrow from Anywhere: Pseudo Multi-modal Object Detection in Thermal Imagery. Chaitanya et al , 2019
- Every Feature Counts: An Improved One-Stage Detector in Thermal Imagery. Zhu et al ,2019
- Visible to infrared transfer learning as a paradigm for accessible real-time object detection and classification in infrared imagery. Hongxu et.al 2024
- Breaking the rain barrier: A novel approach in image processing with AMGR-Net. Zhang et. al 2024.
- CFMW: Cross-modality Fusion Mamba for Robust Object Detection under Adverse Weather. Li et al. 2025

DATASETS

- FLIR ADAS Dataset(Does not have paired RGB and Thermal images)
- M3FD Dataset
- C3I Dataset
- Severe Weather Visible Infrared (SWVI) Dataset.



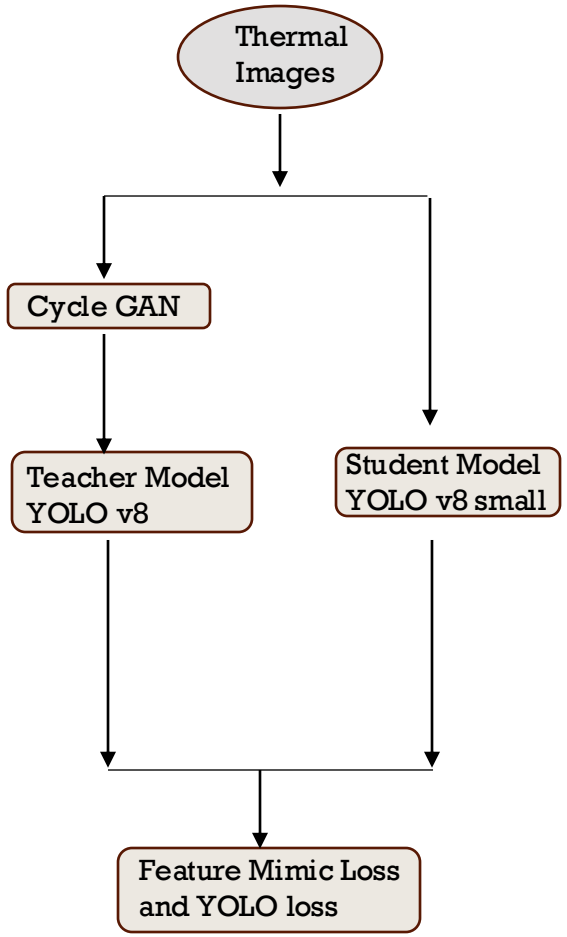


Fig: Data Distillation

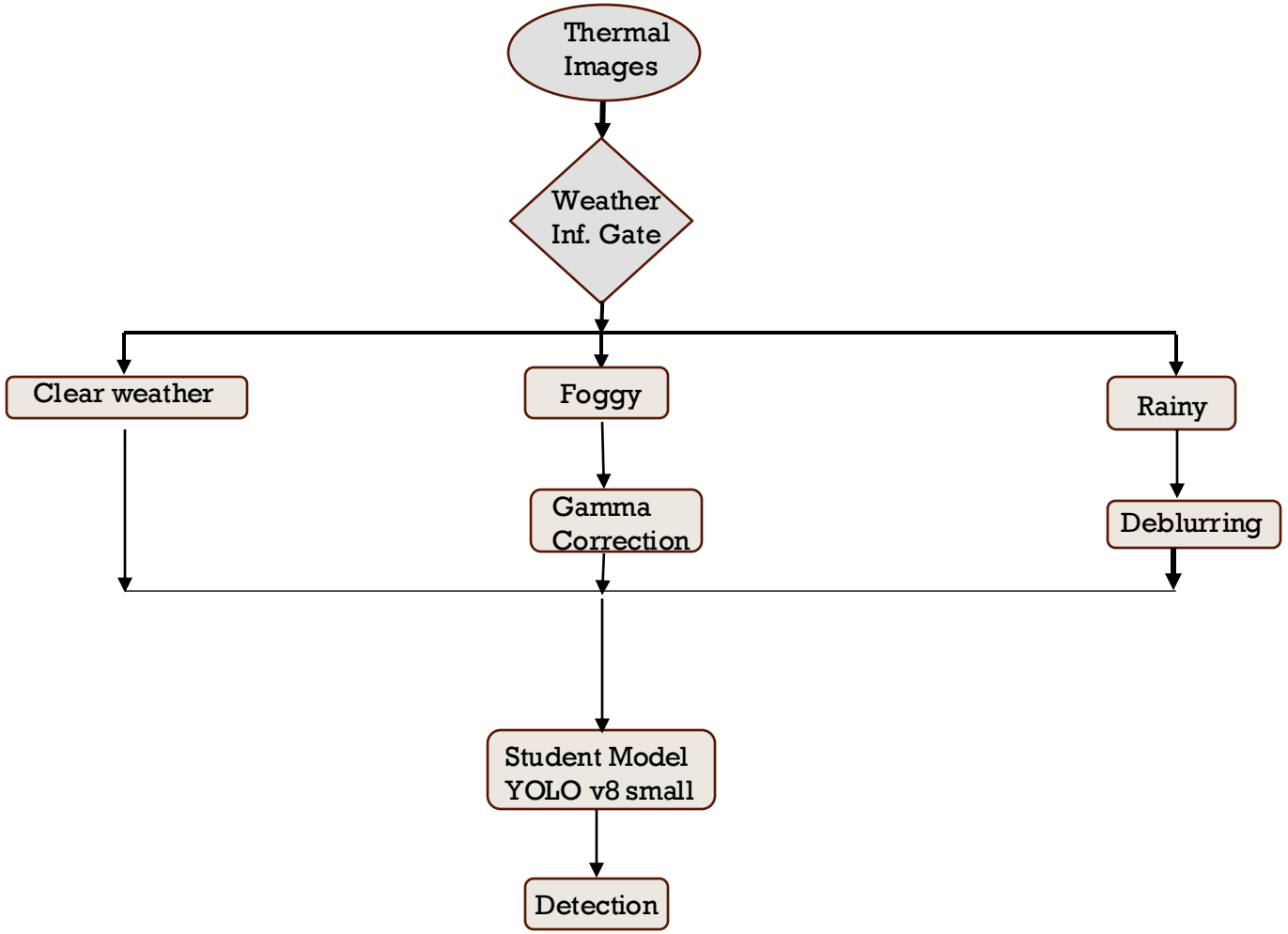


Fig: Forward Inference

THANK YOU

